الجمهورية اللبنانية
وزارة التربية والتعليم العالي
المديرية العامة للتعليم المهني والتقني
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الامتحانات الرسمية لشهادة الامتياز الفني
لدورة عام 2016 الأولى

الاسم والشهرة:----------

رقم المرشح :----------

الاختصاص والرمز : الانظمة والشبكات (23204)
المادة : أنظمة استثمار الجهاز
المستندات المسموح بها : لا شيء

المعدل: (14)
المدة : ساعتان

Treat all the followings questions:

Question I : Explain how the UNIX operating system uses the data structure to manage directories and files. (2pts)

Question II : What is the difference between the two types of **block** and **character** devices? Give an example for each type. (2pts)

Question III : (2pts)

a) Define the deadlock.
b) Give an example illustrating the deadlock problem.

Question IV : (2pts)

a) Draw the structure type of the operating system by specifying its different layers.
b) What is the role of the kernel in the operating system.

Question V : (2pts)

a) How to prevent a high priority process to run indefinitely?
b) How to promote the implementation of the process performing many I/O, so that they don't clutter the memory during their wait?

Question VI : In a multiprogramming machine, we simultaneously performs three processes: (2pts)

P1: I, I, E, I, S, I, I, I, I, I
P2: I, E, I, S, I
P3: I, E, I, I, S, I

Where **I** is a statement which the execution time is 0.25ms, E and S are respectively the input and output operations lasting each 1ms.
a) - Draw a diagram to calculate, for each process, its execution time.
b) - Calculate for the processor its duration of activity and inactivity.

Question VII : Consider the following set of process, with the duration of central unit cycle in ms: (3pts)

Process	Cycle duration	priority
P1	7	3
P2	3	1
P3	2	3
P4	1	4
P5	4	2

It is assumed that the process arrived in order P1, P2, P3, P4, and P5 at time 0.
a)- Draw 4 Gantt charts illustrating the implementation of these processes by using the FCFS, SJF algorithms, a priority without requisition (the smaller number indicates highest priority), and RR (with quantum is 2).
b) - What is the average time of refund (treatment) of a process for each of the scheduling algorithms.

Question VIII : Given P1 and P2 two processes running simultaneously on a multiprogramming machine. (3 pts)

```
P1 ()
Begin
        Var i : char
        For i: = 'A' to 'Z' do write (i);
        Writeln;
        For I: = '0' to '9' do write (i);
        Writeln;
End;
```

```
P2 ()
Begin
        var i : char;
        for i: = 'a' to 'z' do write (i);
        writeln;
End;
```

Using **semaphores**, change these two processes so that they appear in the order, the uppercase letters, lowercase letters and then numbers.

A B C……..Z
a b c……z
0 1 2 ……...9

Question IX : *Given the following table pages* :

Virtual Page	physical Page
0	4
1	6
2	8
3	9
4	12
5	1

Knowing that the virtual and physical pages are 1K bytes, what is the memory address corresponding to each of the following virtual addresses encoded in hexadecimal: **12AB** and **13ED**. (2 pts)
